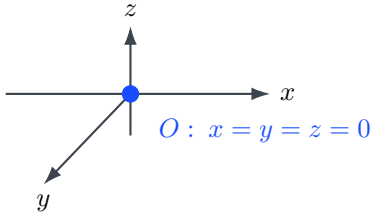
# 约束几个独立方向，就需要几个 $\delta$ 因子

原点点电荷 $Q$

$\rho_Q = Q\delta(x)\delta(y)\delta(z)$

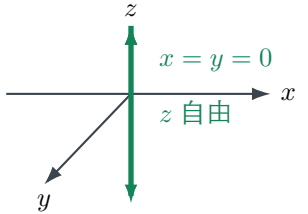



3 个约束；对体积积分得 $Q$

$[\rho_Q] = \mathrm{C\,m^{-3}}$

整条 $z$ 轴：线密度 $\lambda$

$\rho_\lambda = \lambda\delta(x)\delta(y)$

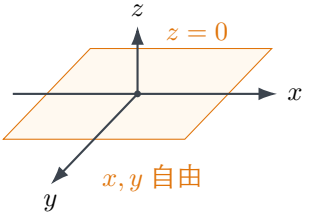



2 个约束；对横截面积分得 $\lambda$

$[\lambda] = \mathrm{C\,m^{-1}}$

整个 $z = 0$ 面：面密度 $\sigma$

$\rho_\sigma = \sigma\delta(z)$




1 个约束；沿法向积分得 $\sigma$

$[\sigma] = \mathrm{C\,m^{-2}}$

线段和面片只表示无限源的局部；三种 $\rho$ 均为三维体密度，单位均为 $\mathrm{C\,m^{-3}}$。